What Helps a Small CNN Generalize? A 2 × 2 Ablation of Batch Normalization and Data Augmentation

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Code: github.com/Marclianjie/data-science-portfolio/tree/main/cnn-ablation-dogs-vs-cats

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1. Question

Batch normalization (BN) and data augmentation (DA) are two standard ways to help a small convolutional network train and generalize, but they act through different mechanisms: BN changes how the network is optimized, while DA changes the distribution of training images. This project separates their individual and joint effects with a controlled 2 × 2 ablation in which every other part of the pipeline is held fixed.

2. Setup

- **Data.** Dogs vs. Cats: 15,000 training images (7,500 per class) and 10,000 validation images (5,000 per class), resized to 128 × 128.
- **Model.** Three convolutional layers (16, 32 and 64 channels; 5 × 5 stride 2, 3 × 3 stride 2, 3 × 3 stride 1) with two max-pooling layers, followed by a 2304 → 128 fully connected layer (ReLU, dropout 0.5) and a two-class output; Kaiming initialization. The network has 319,650 parameters without BN and 319,874 with BN (BN adds 224, or 0.07%); the counts were checked by hand against the model summary.
- **Factors.** BN after each convolutional layer (on/off) crossed with DA (random crop with 8-pixel padding plus random horizontal flip; on/off). Validation images are never augmented.
- **Training.** Adam with learning rate 10^{-3}, batch size 128, cross-entropy loss, 30 epochs. All four arms use the same random seed for the initial weights and the same seeded batch order, so they differ only in the two factors.

3. Results

Configuration	Best validation accuracy (epoch)	Final accuracy (epoch 30)	Final training loss
Baseline	82.21% (11)	80.34%	0.048
+ BN	82.91% (26)	82.51%	0.124
+ DA	87.64% (30)	**87.64%**	0.281
+ BN + DA	86.36% (26)	82.47%	0.321

Table 1. Validation accuracy and training loss for the four configurations.

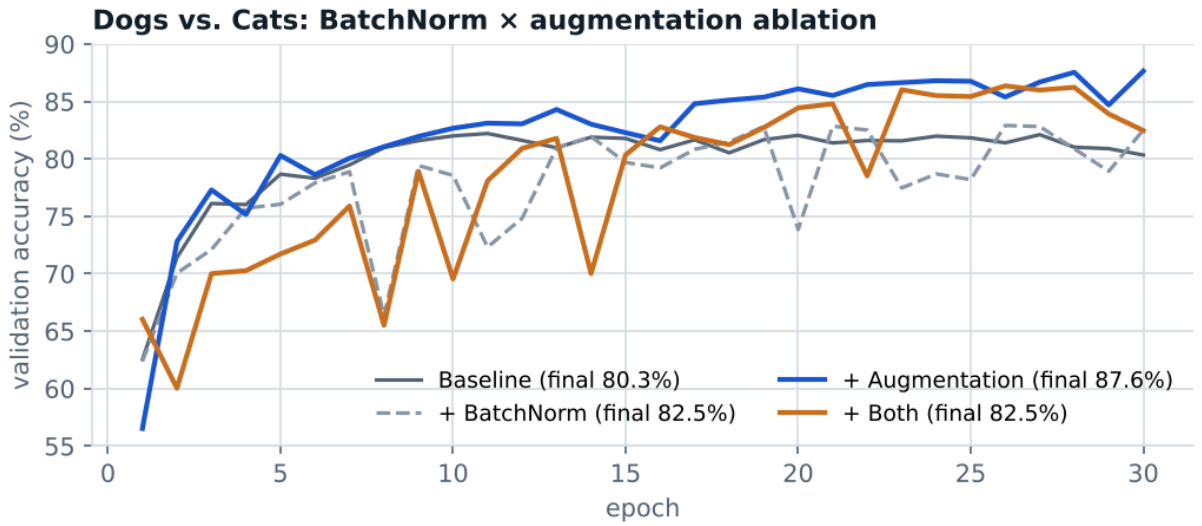


Figure 1. Validation accuracy by epoch for the four configurations.

4. Findings

- **Augmentation gives the largest gain.** Final validation accuracy rises by 7.3 points over the baseline (87.64% vs. 80.34%). Without augmentation the baseline's training loss keeps falling, to 0.048 by epoch 30, while its validation accuracy peaks at epoch 11 and then declines: the network increasingly fits the training images rather than the task. With augmentation, training loss stays near 0.28 and validation accuracy is still rising at epoch 30.
- **Batch normalization mainly steadies training.** BN adds 2.2 points in final accuracy (82.51% vs. 80.34%) but only 0.7 points at the best epoch (82.91% vs. 82.21%), so its benefit here comes from a steadier late phase of training rather than a higher peak.
- **The two did not stack.** BN + DA reached a best of 86.36% but ended at 82.47%, below augmentation alone.

5. Limitations

- Each arm was run with one seed. The BN × DA interaction, and the final-epoch value for BN + DA in particular, should be re-checked over several seeds before any conclusion is drawn from it.
- Best-epoch accuracies are chosen on the same validation split they are reported on and are therefore optimistic; final-epoch accuracy is the fairer comparison, which is why both are shown.
- The learning rate and the 30-epoch budget were fixed. The augmented arm had not converged, so longer training could widen its lead.